

OSRNet: Occlusion Recovery and Relationship Detection of Irregular-shaped Objects

Qiaokang Liang, *Senior Member, IEEE*, Xiangyu Zhang, Hai Qin, Li Zhou, Weilai Jiang, Wei Sun, Dan Zhang, *Senior Member, IEEE*, and Yaonan Wang

Abstract—Existing computer vision systems have surpassed human capabilities in understanding the visible parts of objects, but there is still a gap in accurately describing the occluded parts of objects. This is particularly evident in the case of occlusion between irregular objects in practical solid waste sorting scenarios, such as robotic waste classification. This paper proposes a two-stage convolutional neural network-based solution for occluded object understanding and reconstruction, introducing the Occlusion Segmentation-Recovery Network (OSRNet). Firstly, we create the first kitchen waste occlusion object dataset (KWOD), addressing the lack of datasets for occlusion-related research in previous works. Secondly, the OSRNet model is trained in two stages: the first stage trains a representative instance segmentation model to obtain the category, position, and visible parts of the occluded objects; in the second stage, we design the Swing convolutional Network (SwingNet) to reconstruct the visible portions of occluded objects into their visually complete forms. The two-stage outputs enable the sorting robot to accurately grasp occluded irregular objects. Experimental results on the public datasets COCOA and KINS, as well as the proposed KWOD dataset, demonstrate that our model achieves mIoU scores of 91.19%, 82.55%, and 96.67%, respectively, in the task of occlusion restoration—outperforming existing methods. In real-world testing, a solid waste sorting robot equipped with our approach was able to accurately grasp occluded and irregularly shaped objects, achieving an average sorting success rate of 85.78%. These results validate the superiority and strong generalization capability of the proposed method.

Index Terms—Solid Waste, Occlusion, Recovery, Robotic sorting.

I. INTRODUCTION

THE recognition and detection of occluded objects is a challenging problem in the field of computer vision [1]–[3]. It refers to the situation where part of an object in an

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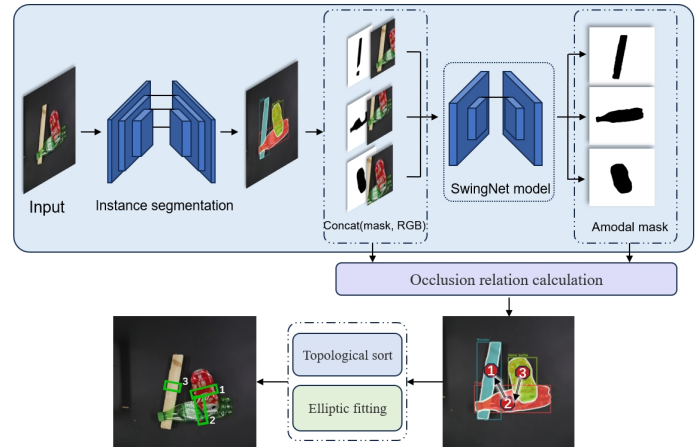


Fig. 1. The input image undergoes instance segmentation, resulting in instance masks for each object, marked with different colors in the image. These instance masks are then separated into independent (visible) instance masks. The mask recovery model is applied to restore the complete shape of each object. The recovered instance masks are fused back into their respective positions in the RGB image to obtain the full shape information of the objects. This complete shape information is then used to generate the grasping boxes. The modal masks and amodal masks are used to infer the occlusion relationships between objects.

image is obstructed by other objects, resulting in incomplete information about the target object. The occlusion problem is prevalent in complex real-world scenarios such as traffic monitoring, crowd analysis, and intelligent sorting. For instance, in the current research [4], [5] hotspot of waste classification, there are numerous cases of occlusion, which is a significant factor contributing to the low classification accuracy.

Waste management is a global challenge, with billions of tons of waste generated daily, requiring significant costs for disposal. According to existing data and projections, from 2020 to 2050, urban solid waste will emit 32 to 35 billion tons of CO₂-equivalent greenhouse gases [6]. The United Nations advocates for zero waste emissions, emphasizing the reuse and recycling of resources as much as possible. In the task of intelligent solid waste sorting, challenges arise due to the wide variety of waste types, significant shape differences, and severe occlusion among objects. Currently, robotic sorting systems face high misgrasp rates due to low recognition accuracy. Therefore, research on visual methods to address the occlusion problem for irregular-shaped targets is of significant research importance and practical application value.

With the development of technology, the integration of artificial intelligence in waste sorting has become a growing

trend. This paper seeks to address the occlusion problem in the kitchen waste sorting process. Our goal is to segment the complete masks of each recyclable object in the image, determine the occlusion relationships between objects, and then allow the sorting robot to perform accurate grabs based on the complete masks and occlusion relationships. In the kitchen waste sorting scenario, cameras are mostly placed upstream on the conveyor belt, while sorting robots are downstream, resulting in an "eye-off-hand" situation. As such, real-time inference for the model is not a critical factor. With the continuous improvement of instance segmentation techniques, segmentation models have become increasingly effective, with faster inference speeds, and can deliver good results in waste sorting applications. For mask recovery, we build upon the PCNet method [7], which provides insights for recovering visible masks into complete masks, thereby addressing the occlusion problem in the kitchen waste sorting process.

As a deep learning-based approach, high-quality data and annotations are essential prerequisites for achieving good model performance. To this end, we have established the Kitchen Waste Occlusion Object Dataset (KWOD) tailored to the practical scenario of kitchen waste sorting on a conveyor belt. The specific details of the dataset will be discussed in Section 3.1. The training process for the proposed model consists of two stages, as shown in Fig. 1. In the first stage, we train an instance segmentation model to segment the visible parts of recyclable objects in the image, obtaining the object's position, category, and visible portions. In the second stage, we combine the labeled visible masks with the instance segmentation masks (after stripping the class information) through mixing. After data augmentation, the RGB images of the objects are used as inputs to the recovery model, training it to generate the complete masks for the occluded objects. Next, we infer the occlusion relationships between objects from the complete masks and visible masks (modal masks), which allows us to determine the order in which objects should be grasped. Finally, the complete masks are used to generate the object grasping positions and the gripper pose. The proposed approach has significant reference value for other occlusion-prone scenarios, such as autonomous driving, robot path planning, and sorting tasks.

The main contribution of this paper is the proposal of a complete framework for occlusion relationship detection and restoration, aiming to reconstruct occluded objects into visually complete forms. The key contributions are summarized as follows:

- 1) We built the first occluded solid waste dataset (KWOD) with complex occlusion conditions, such as multiple occluders, variable object sizes, and high overlap rates—filling a key gap left by prior datasets.
- 2) We proposed an occlusion reasoning model based on instance segmentation, addressing the issue where visible parts of occluded objects were misclassified as separate entities. We also introduced a restoration model SwingNet to reconstruct occluded items into visually complete forms.
- 3) We designed a novel Atten-Res-Block, which integrates ResNet blocks with spatial and channel attention me-

chanisms. Combined with our proposed Composite Occlusion Task Loss (COT-Loss) strategy, our model achieved a 96.67% mIoU on the KWOD dataset, outperforming existing methods.

The source code and dataset examples are available at our github repository: <https://github.com/yulle13/OSRNet>

II. RELATED WORKS

A. Instance segmentation

Instance segmentation involves classifying each pixel in an image and distinguishing between different target objects [8], [9]. As a technique that simultaneously solves both instance-level object detection and semantic segmentation problems, it plays a crucial role in helping computers understand visual information [10]. Instance segmentation methods are generally divided into one-stage and two-stage approaches. One-stage methods include the initial FCIS [11], which computes a position-sensitive inside/outside score map and outputs instance masks along with class information. YOLACT [12] extends one-stage object detection models by adding a mask branch, while BlendMask [13] combines top-down and bottom-up approaches, making it more effective for densely packed objects. The well-known YOLO [14] segmentation model has been improved based on YOLACT, maintaining high speed while providing more accurate segmentation results. More recently, state-of-the-art YOLO variants have further advanced the field: YOLOv10 [15] achieves real-time end-to-end object detection by eliminating the need for non-maximum suppression, YOLOE [16] extends the YOLO framework to open-vocabulary and promptable detection for real-time seeing anything, and YOLO-UniOW [17] introduces an efficient universal open-world object detection paradigm capable of detecting both known and unknown categories. Two-stage instance segmentation models first extract candidate regions and then refine and segment these regions. The most representative model is Mask R-CNN [18], which uses a Region Proposal Network (RPN) to generate candidate regions, followed by classification, bounding box regression, and segmentation mask generation. An improvement on this is Cascade Mask R-CNN [19], which employs a cascaded structure to enhance segmentation performance and accuracy, making it suitable for handling complex backgrounds and highly overlapping instances. HTC [20] combines multi-task learning by adding a semantic segmentation branch to the Cascade Mask R-CNN. Currently, most research on instance segmentation in the context of occlusion is conducted on publicly available datasets, such as COCO [21] or hyperspectral images [22]. There is limited application of these methods in practical complex occlusion scenarios, such as solid waste sorting.

B. Amodal Instance Completion

Amodal Instance Completion differs from other typical tasks in that it involves restoring the occluded parts of an object using the given modal mask to obtain the amodal mask. This task mimics human cognition, where the visible portion of an occluded object is imagined to complete the full shape

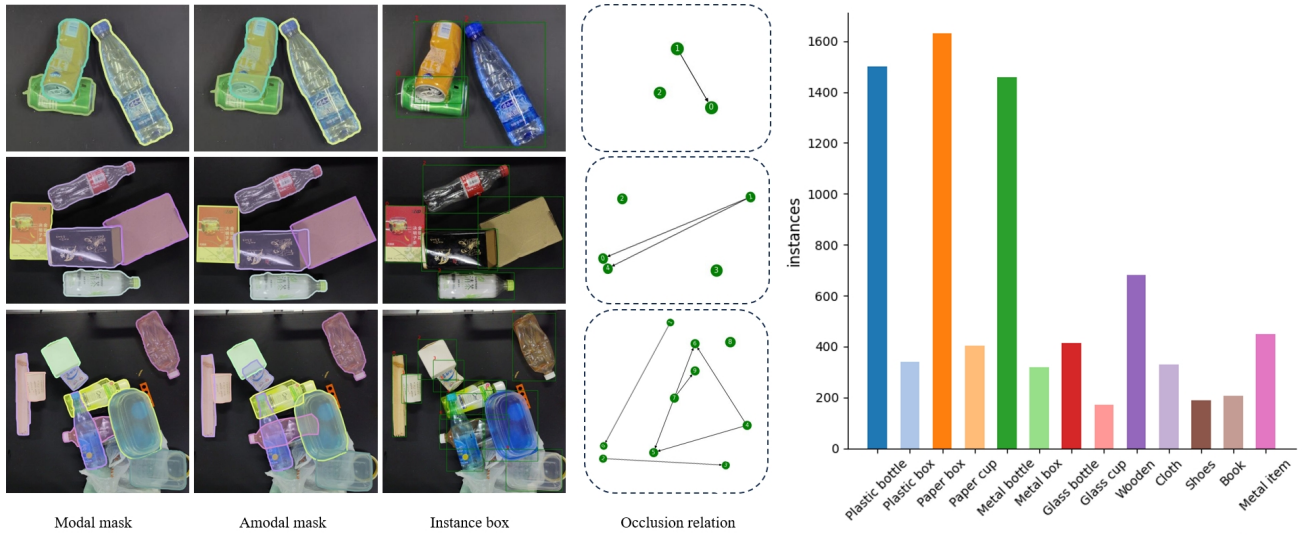


Fig. 2. The Modal Mask represents the visible portion of each recyclable object, while the Amodal Mask depicts the complete shape of the object, including both visible and occluded parts. The Instance Box is generated from the Amodal Mask and provides the bounding box around the object. During annotation, each object in the image is assigned a unique instance ID, and the Occlusion Relationship Graph is formed based on the occlusion relationships derived from these instance IDs. On the right side of the image is the number distribution of each category in the dataset.

of the object [23]. In previous research, Li and Malik [24] first proposed a method to extend the bounding box of a mask along the high heatmap values [25], which increases the occlusion situation in a comprehensive manner. Zhu et al. [26] selected a dataset from the original COCO, creating the COCO Amodal Dataset, which contains 5,000 images and uses AmodalMask as the baseline. Their method trains SharpMask [27] with amodal ground truth. In the field of autonomous driving, Qi et al. [28] created a newly annotated instance segmentation dataset, KINS, from the large-scale KITTI dataset [29] and proposed ASN to improve amodal segmentation performance. Building on this, Zhan [7] introduced the self-supervised model PCNet, which does not rely on complete masks for training but still achieves good restoration results. Khoi [30] further proposed the ASBU model based on PCNet, which utilizes occlusion boundary information to restore masks, bypassing the dependence on occlusion relationship maps used by the PCNet network. Li et al. [31] proposed a Generative Invariant Shape Prior Network (GIN) to simulate human perception, learning basic shapes, and achieving effective results in predicting the full shape of occluded objects in public datasets. However, current research still faces challenges, as heatmap masks and simple mask inputs often lead to distorted restored object shapes, and even adding occlusion boundary information does not improve the model's performance effectively.

C. Other Occlusion Handling

Unlike the approach that first applies instance segmentation followed by mask restoration, Lee and Park [32] utilized a depth camera, combined with their proposed InstaOrderNet and InstaDepthNet networks, to accurately predict the occlusion relationships of objects in an image. In hierarchical methods, Zheng [33] introduced the concept of using pseudo-ground-truth generated from real images. Their proposed CS-

DNet model is based on the idea that objects detected as fully visible in one layer are actually occluders of objects in the subsequent layers. Ke et al. [34] modeled the image as composed of two overlapping layers and used a dual-layer structure to explicitly model occlusion relationships, decoupling the boundaries of occluded and occluding instances. The DOC-VTON method [35] employs semantic guidance and pose priors, using a copy-paste technique to selectively blend complex textures with human body parts, and a generative module is used to synthesize the final image, with a focus on occlusion removal. In contrast to the methods mentioned above, we propose a different approach where the color image is fused with the visible mask. By designing a dedicated network, we aim to extract deeper information about the occluded objects, enabling us to restore the true appearance of occluded objects with high quality.

III. METHODOLOGY

A. Dataset Description

In the field of computer vision, the importance of datasets is widely recognized, as they provide the necessary foundation for training, debugging, and comparing different algorithms and frameworks. Therefore, constructing a high-quality dataset

TABLE I
COMPARISON OF ATTRIBUTES OF REPRESENTATIVE DATASETS IN THE OCCLUSION DOMAIN.

Dataset	Scale	Categories	Data Source	Occlusion Level		
				Mild	Moderate	Severe
COCOA	5K	1000	Real-world scenes	✓	✓	
COCOA-cls	3.5K	5	Real-world scenes	✓	✓	
instaOrder	101K	80	Real-world scenes	✓	✓	
KINS	15K	8	Real-world scenes	✓	✓	
D2S Amodal	2K	60	Synthetic	✓		
KWOD	2.1K	13	Industrial scenes	✓	✓	✓

is of significant value. In the domain of amodal datasets, there are both real-world datasets, such as COCOA [26], COCOA-cl [36], InstaOrder [32], and KINS [28], and synthetic datasets like D2S Amodal [36]. The former mainly consist of occlusion scenarios and backgrounds that are relatively simple and commonly found in daily life, while the latter are synthetically generated with artificially controlled occlusion complexity. At present, most occlusion-related datasets remain at the research stage. To address this limitation, we collected thousands of images from a kitchen waste sorting production line characterized by complex occlusion patterns, and constructed a dataset tailored for practical industrial applications. This provides a valuable reference for applying vision-based and deep learning techniques to real-world engineering problems involving complex occlusions. These existing datasets served as references in the development of our kitchen waste occlusion dataset (KWOD). A comparison of their attributes is presented in Table I.

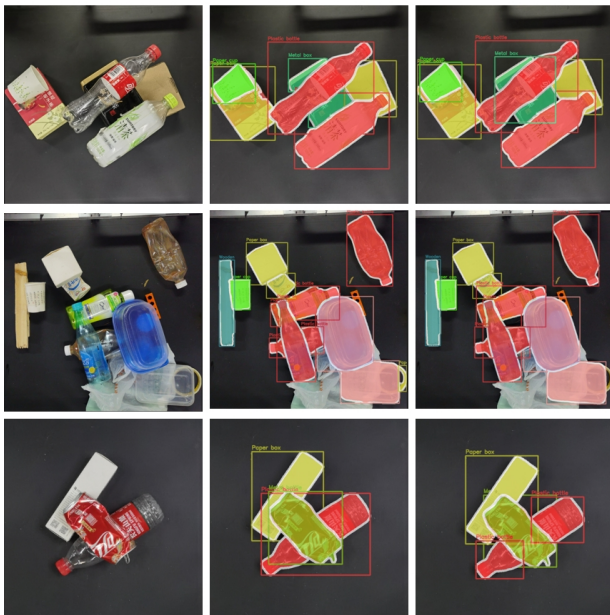


Fig. 3. Instance segmentation splits the objects in the image into individual instances. In our data processing section, we made modifications to the instance segmentation model to enable it to recognize objects with occlusions in the middle as the same object. The image on the right shows the comparison between the original and modified results.

The goal of creating the KWOD dataset is to address the challenges of occlusion and stacking in the garbage sorting domain, providing accurate grasping positions and postures for robotic sorting, thereby improving sorting efficiency. In our dataset, annotations are divided into three main categories, (1) Visible Mask Annotations: Marking the visible portions of each object in the image. (2) Complete Mask Annotations: Labeling the full object shape, including both visible and occluded parts and (3) Occlusion Relationship Annotations: Denoting the occlusion relationships between objects in the image, showing how one object occludes another. An example of the annotations is shown in Fig. 2, where these three types of annotations help train models to not only recognize

objects in cluttered scenes but also predict their full shapes and relationships.

Due to variations in people's eating habits and seasonal changes, the types of waste generated can differ at different times. To simulate different time periods on a garbage sorting conveyor, we collected 2,117 images of recyclable waste. Fig. 2 shows the statistical results of the recyclable objects. We categorized the recyclable items into 13 common types, with a total of 10,356 instances. The dataset was divided into training and validation sets using an 8:2 ratio.

For occlusion relationships, in the KWOD dataset, we categorized the images based on the complexity of the occlusions as follows: Two-object occlusion: 200 images. Mild occlusion: 1,000 images. Moderate occlusion: 500 images. Severe occlusion: 300 images. Definitions for each occlusion level are as follows. Mild occlusion: The background is relatively clean, and the occlusion of objects does not exceed 25%. The occlusion depth is no more than 3 layers. For example, if A blocks B, and B blocks C, the occlusion depth for C is 3. Moderate occlusion: The background is somewhat cluttered, with no occlusion in the foreground. The occlusion of objects does not exceed 50%, and the number of objects in the image does not exceed 7. Severe occlusion: The background is messy, and non-target foreground elements (e.g., plastic bags, transparent films) cause occlusion. The occlusion of objects exceeds 50%, or there are more than 7 objects in the image.

B. Segmentation-Completion Network Model Introduction

The central task of this paper is to solve the occlusion problem in kitchen waste sorting using visual methods. We extend the previously established approach of first performing instance segmentation and then applying mask completion, as shown in Fig. 3. Specifically, when given a color image, we first apply instance segmentation to obtain the visible masks of the various objects in the image. Next, the designed mask completion model (SwingNet) is used to restore the masks, providing the full, unoccluded shape of the objects. By leveraging both the visible masks and the completed masks (i.e., amodal masks) for pairs of objects, we can determine the occluder and the occluded object.

In the instance segmentation part, we selected several state-of-the-art (SOTA) instance segmentation models as baseline methods for comparison. In previous instance segmentation models, whether using a bottom-up approach (semantic segmentation followed by clustering) or a top-down approach (object detection followed by semantic segmentation), there was an issue when the target object was fragmented into multiple disconnected regions. This would lead to the object being recognized as multiple separate objects or only part of it being detected. As shown in Fig. 3, we improved the model's recognition of occluded objects that are split into multiple disconnected regions by treating them as a single object. The advantage of this approach is that it allows for the maximum extraction of visible masks. Regarding previous research on restoring visible masks to complete masks, such as PCNet [7], ASBU [30], and others, these models employed weakly supervised training, where only the visible mask of

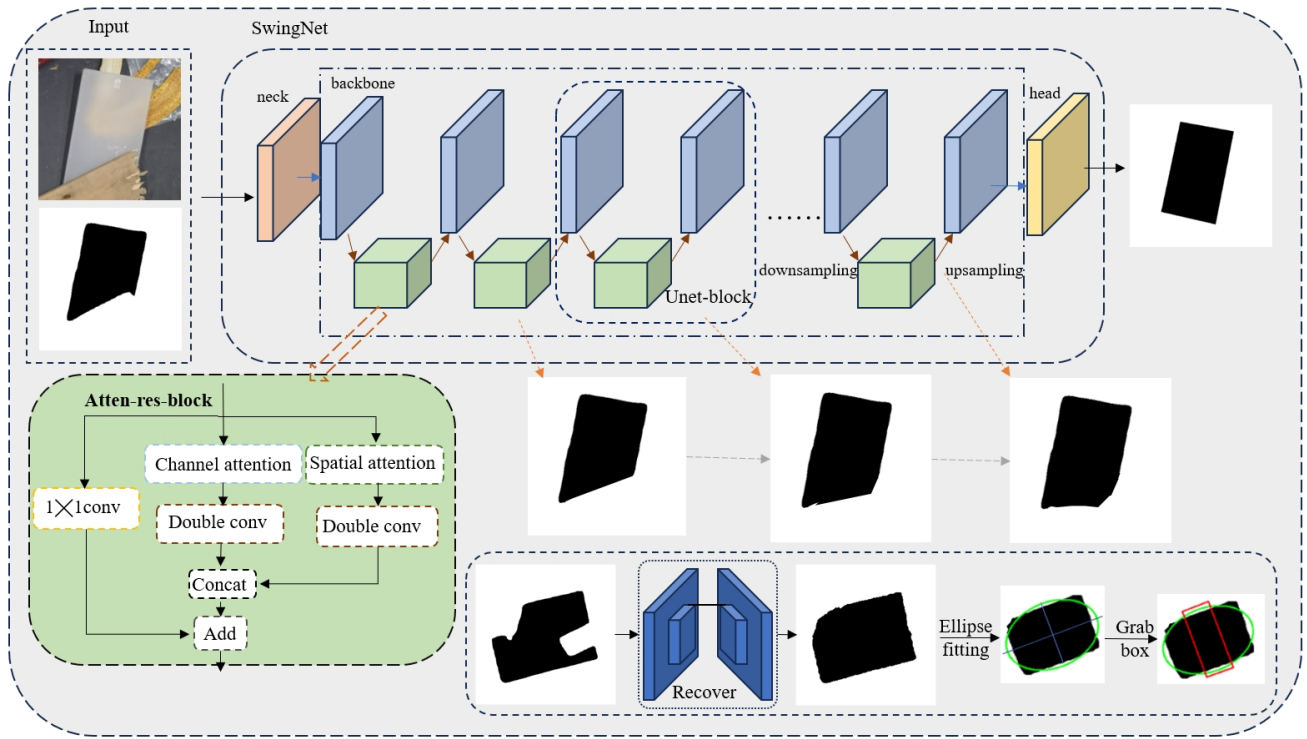


Fig. 4. The proposed SwingNet consists of three main components: neck, backbone, and head. The backbone is composed of a series of simplified Unet modules. The training input to the model includes the RGB image of the object and its visible mask. The bottom right corner of the diagram shows the process of generating the grab frame.

the object was input without incorporating the RGB image, and real complete masks were not used during training. This often led to severe overfitting and distortion. It is analogous to a human who has never seen the complete shape of an object being unable to imagine and restore its full shape when occluded. To address this issue, we conducted extensive experiments demonstrating that fully supervised training for mask restoration (amodal completion) yields better practical results, as detailed in the experimental section (Section 4).

The SwingNet model proposed in this paper is used for mask completion, and its structure is shown in Fig. 4. As illustrated in Fig. 4, SwingNet consists of three main components: a *neck* (input processing), a *backbone* (core feature extraction), and a *head* (output prediction). The backbone is the central component of SwingNet and is composed of a cascade of L simplified Unet-blocks (where $L=3$ in our final model). Each Unet-block is a self-contained encoder-decoder module derived from the Unet architecture [37], serving as the fundamental building block of the backbone rather than a connection module between separate components. By concatenating multiple Unet-blocks, a “wave-like” structure is formed, which constitutes SwingNet. The name “Swing” is inspired by the oscillatory motion pattern of the feature maps through the network: as data passes through each Unet-block, the spatial resolution first contracts (downsampling) and then expands (upsampling), resembling a pendulum-like swinging motion. When multiple such blocks are cascaded, the feature resolution repeatedly swings between compressed and expanded states, forming a characteristic oscillating trajectory. This swing mechanism is tightly coupled with the core

components of the network: (1) each Unet-block acts as one “swing cycle,” progressively refining the mask prediction by alternating between global context extraction (at low resolution) and local detail recovery (at high resolution); (2) the Atten-Res-Blocks embedded within each swing cycle serve as the “pivot points,” using channel and spatial attention to selectively amplify the most informative features at each resolution transition; and (3) the skip connections within each block act as “restoring forces,” ensuring that fine-grained spatial information is preserved across each swing cycle. Together, these components create a rhythmic, progressive refinement process where each swing cycle contributes incrementally to the final mask completion. Each Unet-block consists of downsampling and upsampling operations. During downsampling, the feature map size is reduced while the number of channels increases; during upsampling, the feature map size is increased while the number of channels is reduced. The upsampled feature map is then fused with the input feature map. Each Unet-block is responsible for part of the mask completion task, and the intensity of this completion task is determined by the total number of modules. After passing through multiple such blocks, the mask is progressively completed. In the downsampling and upsampling operations within each module, we incorporate Atten-res-blocks, which combine Res-blocks with an attention mechanism. The Channel Attention weighs the channels of the feature map to enhance the discriminative features, while Spatial Attention focuses on key spatial features of the image. The addition of attention mechanisms reduces the reliance on depth and width, achieving a lightweight design for the model,

echoing recent advances in efficient network architecture design [38], [39].

C. COT- Loss: Amodal Completion Combined Loss

The loss function is a critical metric for measuring the difference between the model's predictions and the ground truth labels, and it plays a vital role in guiding the model's training process. Selecting an appropriate loss function is crucial for the model's performance. Whether for single-task or multi-task learning, it is common to encounter situations where multiple loss functions are involved. A straightforward approach is to apply a linear combination to weight each of the individual loss functions.

$$L_{total} = \sum_{i=0}^n \omega_i L_i \quad (1)$$

Previous works such as Uncertainty Weighting [40] adopt a multi-task loss function based on Gaussian likelihood maximization to handle uncertainty, while GradNorm [41] dynamically adjusts the gradient values of each task through gradient normalization, among other strategies. Given that the goal of this paper is to restore masks, we propose a combined loss function, COT-Loss, specifically designed for the occlusion task. This loss function consists of three components: Cross-Entropy Loss, Dice Loss, and Invisible Loss. Cross-Entropy Loss serves as a standard loss function and is the cornerstone for the restoration task. Dice Loss helps the model learn more accurate boundary information, improves small object detection, and effectively mitigates the class imbalance issue. Invisible Loss is the third component of COT-Loss. It is not a fundamentally different type of loss function; rather, it shares exactly the same cross-entropy formulation as the standard Cross-Entropy Loss but is restricted to a specific spatial region. Concretely, while the standard Cross-Entropy Loss computes the pixel-wise error over the *entire* predicted mask (both visible and invisible regions), the Invisible Loss computes the pixel-wise cross-entropy error over only the *invisible* (occluded) region pixels, which are identified by the difference between the amodal mask and the modal mask. We therefore consistently refer to it as a *region-restricted cross-entropy* term throughout this paper. Formally, let M_a denote the ground-truth amodal mask and M_m denote the modal (visible) mask; the invisible region is defined as $R_{inv} = M_a \setminus M_m$. The Invisible Loss is then computed as:

$$L_{inv} = -\frac{1}{|R_{inv}|} \sum_{p \in R_{inv}} [y_p \log(\hat{y}_p) + (1 - y_p) \log(1 - \hat{y}_p)] \quad (2)$$

where y_p and $\hat{y}_p$ are the ground-truth and predicted values at pixel p , respectively. By isolating the invisible region, this loss term forces the model to allocate more learning capacity specifically to the occluded parts that need to be reconstructed, rather than being dominated by the easier-to-predict visible regions. The weight design for the loss functions in this task follows a dynamic adaptive strategy with incremental enhancement. Specifically, during the initial stages of model training, we only use Cross-Entropy Loss and Dice Loss, and the dynamic weight calculation formula is as follows:

$$\omega_i = \frac{\sum_{m=0}^n L_m (m \neq i)}{(n-1) \sum_{j=0}^n L_j} \quad (3)$$

The loss function values are denoted as L_i for each individual loss, and the sum of their weights $\sum_{m=0}^n \omega_i$ is equal to 1. This formula is used to determine the relative weights between different loss functions, with the assumption that all loss functions operate on the same scale. In the mid-training phase, when L_{total} reaches 10% of the initial value, the invisible loss is added as an additive enhancement to the calculation of L_{total} . The choice of the 10% threshold was determined through systematic experimental validation. We conducted experiments with different threshold values (5%, 10%, 15%, and 20%) on the KWOD validation set, measuring the resulting inv-mIoU: 5% yielded 71.83% (the invisible loss was introduced too early, before the model had sufficiently learned the visible region mapping, causing training instability); 10% achieved the best result of 74.31%; 15% produced 73.12% (slightly delayed introduction reduced the benefit); and 20% gave 72.05% (the model had already converged on visible regions and had limited capacity to shift focus). The 10% threshold represents the optimal balance point: at this stage, the model's mIoU on visible areas consistently exceeds 80%, indicating that the basic alignment between predictions and ground truth has been established, while sufficient training budget remains for the model to effectively learn the invisible region reconstruction. Our strategy is based on the observation that after a period of training, when L_{total} reaches approximately 10% of its initial value, the model has achieved an alignment between predictions and ground truth labels. At this point, the visible part of the mask is treated as input data, which is relatively straightforward to map. The model, however, needs to focus more on the reconstruction of the occluded (invisible) parts of the object. The invisible loss function specifically computes the pixel-wise distance between the predicted and true labels for the invisible regions, which exactly satisfies this need. Since the cross-entropy loss already includes a component that calculates the pixel-wise distance for the invisible parts, once the invisible loss function is integrated into the dynamic weight calculation, the optimization direction of the entire model will shift more towards minimizing the pixel-wise errors in the invisible mask. If the invisible loss is incorporated into the dynamic weight calculation from the very beginning, it would lead to an imbalance in the weight magnitudes, causing convergence issues and complicating the model's training process.

D. Post-processing

After the mask completion model, we can obtain the complete mask (amodal mask) for the objects in the image. To obtain the occlusion relationship graph, we need to compare the modal and amodal masks of all objects in a given image. We define (m_i, a_i) and (m_j, a_j) as the modal and amodal masks for two objects i and j , respectively, where m denotes the modal mask and a denotes the amodal mask. The occlusion relationship between the two objects is stored in $O_{(i,j)}$, and it is calculated using the following formula:

$$O_{(i,j)} = \begin{cases} 1, & \text{if } |m_i \cap a_j| > |m_j \cap a_i| \\ 0, & \text{if } |m_i \cap a_j| = |m_j \cap a_i| \\ -1, & \text{otherwise} \end{cases} \quad (4)$$

When $O_{(i,j)}=1$, it indicates that object i occludes object j . On the other hand, if the modal and amodal masks of two objects do not intersect at all, there is no occlusion relationship, and $O_{(i,j)}=0$. Once the occlusion relationship graph $O_{(i,j)}$ is obtained, we construct a directed graph and perform topological sorting to generate the grasping order of the objects in the image.

In the robotic sorting process, what the robot executes are individual grasping boxes. As shown in Fig. 4, Grasp poses are typically defined by an object's position and the orientation angle relative to the robot's end-effector. OSRNet, however, produces a full binary mask for each object, which cannot be used directly for grasping; we must first convert the mask into a grasp rectangle. Empirically, the gripper is most effective when approaching perpendicular to the object's long axis, so we compute that axis via the Minimum Bounding Rectangle (MBR). Given a mask image, we extract its largest contour and denote its point set as $P = \{p_1, p_1, \dots, p_n\}$, $p_i = (x_i, y_i) \in \mathbb{R}^2$, with convex hull $\text{CH}(P)$. For any rotation angle Θ , we apply the corresponding rotation transform to obtain $P' = \{p'_1, p'_1, \dots, p'_n\}$. In this rotated coordinate frame, the horizontal and vertical projection lengths are:

$$w(\theta) = \max_{(x', y') \in P'} (x') - \min_{(x', y') \in P'} (x') \quad (5)$$

$$h(\theta) = \max_{(x', y') \in P'} (y') - \min_{(x', y') \in P'} (y') \quad (6)$$

Thus, the rectangle's area is given by $A(\theta) = w(\theta) \cdot h(\theta)$, and the optimal rotation angle is chosen as:

$$\theta^* = \arg \min_{\theta} A(\theta) \quad (7)$$

The rectangle at θ^* is therefore the minimum bounding rectangle of the point set P . From its four vertices, one can then compute the rectangle's geometric parameters—namely its center coordinates, width, and height.

The grasping box generated from the object's complete shape and the grasping sequence of the objects in the image enable the robot to perform accurate and efficient grasping of recyclable waste, thus achieving high-precision object sorting.

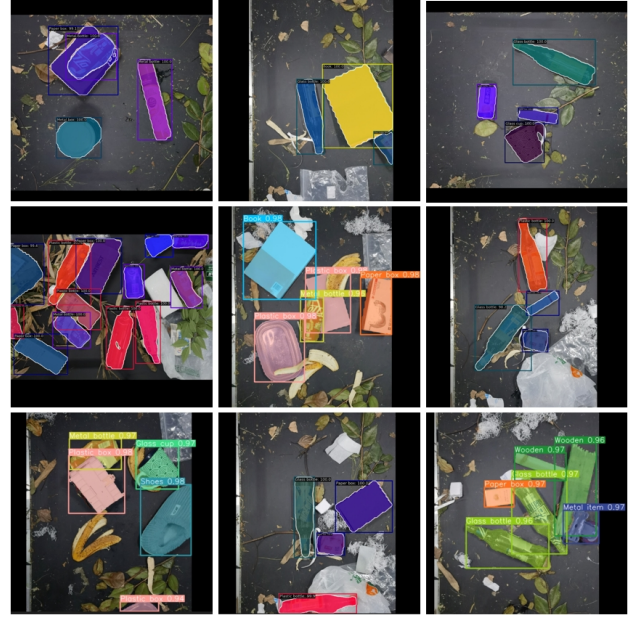


Fig. 5. Predicted results of instance segmentation models on the KWOD validation dataset.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

We evaluate our model on the self-built KWOD dataset, as well as two public datasets: KINS [28] and COCOA [26]. The KINS and COCOA datasets were derived from the MSCOCO [29] and KITTI [42] datasets, respectively. The COCOA dataset consists of 2,500, 1,323, and 1,250 images for training, validation, and testing, respectively. It contains a total of 2,140 object categories, which are divided into two superclasses: "items" (e.g., sky, grass, sea) and "things" (e.g., dogs, cats, people). The images in COCOA mostly contain densely packed objects that are heavily occluded in cluttered scenes. On the other hand, KINS is a large-scale traffic dataset composed of 7,474 training images and 7,517 testing images. The object categories in KINS include 7 classes: cyclists, pedestrians, cars, trams, trucks, vans, and miscellaneous vehicles. The scenes in KINS are less cluttered compared to those in COCOA, and occlusions, when present, are mainly caused by other objects. Both datasets provide ground-truth amodal masks, which allows for a direct comparison of model performance across different methods.

A. Instance Segmentation Experiment

The instance segmentation model serves as the first stage

TABLE II
SEGMENTATION EVALUATION RESULTS OF DIFFERENT MODELS ON THE VALIDATION SETS OF KINS AND KWOD.

Datasets	Model	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ₁	AR ₁₀	AR ₁₀₀	AR _S	AR _M	AR _L
KINS-val	Mask-Rcnn	22.1	39.9	22.1	7.3	31.7	55.0	13.5	28.0	28.9	12.7	38.1	59.6
	Cascade Mask Rcnn	23.0	40.1	23.8	7.8	32.1	56.7	13.4	27.8	28.7	12.5	37.2	60.2
	Solo	17.6	31.4	17.7	3.4	27.3	55.7	11.8	23.7	24.2	6.6	34.1	60.9
	Yolact	12.5	26.5	9.3	1.8	17.7	43.2	9.2	18.8	19.3	5.4	26.3	50.5
KWOD-val	Mask-Rcnn	82.6	93.8	89.2	46.6	52.3	84.6	72.6	87.8	87.8	46.7	55.9	89.7
	Cascade Mask Rcnn	84.4	93.5	90.4	43.4	54.3	86.3	73.3	88.4	88.4	43.3	57.0	90.2
	Solo	82.6	92.4	88.5	33.5	51.3	84.7	71.4	86.9	87.0	36.7	54.7	89.2
	Yolact	73.0	90.6	80.3	4.5	37.9	75.3	66.9	79.9	79.9	13.3	49.3	81.9

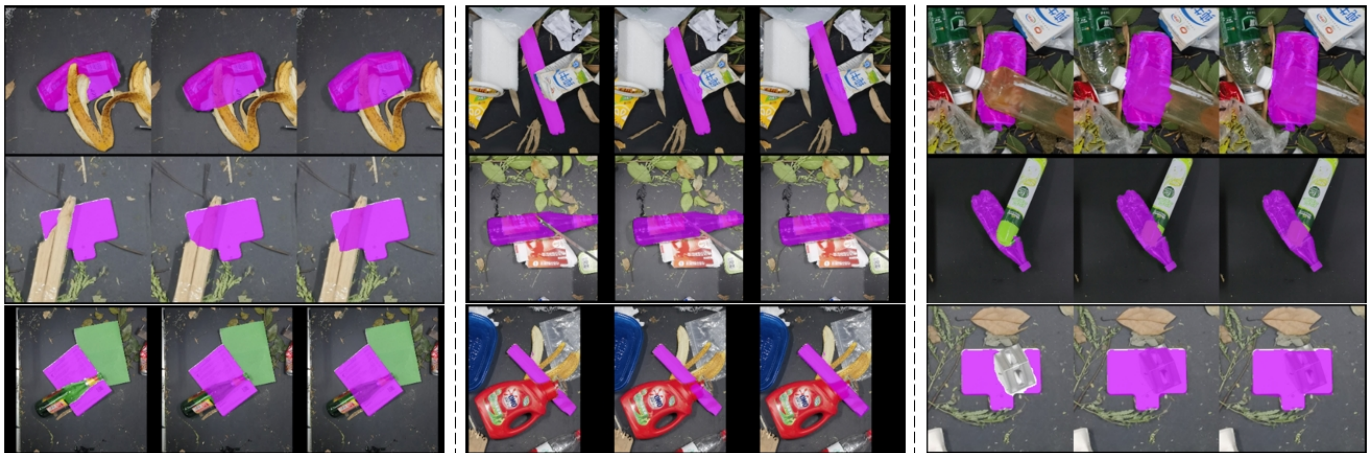


Fig. 6. Mask completion results of SwingNet on the KWOD dataset. Each set of three images shows (from left to right): the modal mask, the amodal mask predicted by the mask completion model, and the ground truth amodal mask.

TABLE III
TEST RESULTS OF FIVE RESTORATION MODELS ON COCOA, KINS, AND KWOD.

Model	Supervision	COCOA-val	COCOA-test	KINS-test		KWOD-val	
		mIoU	mIoU	mIoU	Inv-mIoU	mIoU	Inv-mIoU
PCNet-m [7]	Self-sup.	81.35	86.63	94.76	59.24	86.07	58.08
C2F-Seg [43]	Fully-sup.	80.28	86.42	87.89	57.60	85.21	61.59
ASBU [30]	Weakly-sup.	84.22	89.87	94.83	64.41	91.13	65.21
BCNet [44]	Fully-sup.	82.42	90.07	93.53	62.21	92.03	66.72
SwingNet	Fully-sup.	89.44	91.19	94.22	82.55	96.67	74.31

of our framework, and the quality of its segmentation directly affects the performance of the subsequent restoration model. Since the COCOA dataset contains 2140 categories with highly imbalanced class distributions, it is not particularly meaningful to conduct instance segmentation experiments on it. Therefore, in this paper, we selected the publicly available KINS dataset and our custom-built KWOD dataset to train and evaluate four classic instance segmentation models: Mask R-CNN [18], Cascade Mask R-CNN [19], SOLO [45], and YOLACT [12]. The experimental results are shown in Table II. We applied the data processing improvements described in Section 3 to these four models, enabling them to better recognize objects whose visible parts have been split due to occlusion. Fig. 5 shows the performance of the instance segmentation models on the KWOD dataset. We used the COCO segmentation evaluation metrics, and the backbone of all four models is ResNet-50. The implementations of these models were provided in the MMDetection [46] toolbox. All experiments were run on a PC equipped with two RTX 3090 GPUs and 128 GB RAM.

Table II presents the experimental results of different segmentation models on the validation sets of KINS and KWOD, evaluated using the COCO segmentation metrics. Cascade Mask R-CNN leads in most metrics on both the KINS and KWOD datasets, with a particularly noticeable performance improvement on the KWOD dataset. Mask R-CNN follows closely, which is understandable, as Cascade Mask R-CNN is an improvement over Mask R-CNN. SOLO and YOLACT perform relatively weaker, but their fast inference speed makes them suitable for real-time applications where low latency is

required.

From the experimental results, we can observe that the evaluation metrics on our self-built KWOD dataset are generally higher than those on the KINS dataset. The main reasons for this are as follows: the KWOD dataset contains fewer images compared to KINS, the objects are placed on a single conveyor belt background, and the lighting conditions are stable. The higher segmentation quality in the KWOD dataset provides better segmentation masks for subsequent mask completion, thus improving the performance of the restoration model.

Traditional instance segmentation models often struggle with occluded objects due to missing information in the occluded regions, resulting in incomplete mask predictions. This limitation negatively impacts subsequent semantic understanding and downstream tasks. In this work, we address this challenge by constructing the KWOD dataset, which features complex occlusion relationships, and by employing powerful multi-stage feature extraction models such as Cascade Mask R-CNN. This approach significantly improves segmentation accuracy, particularly in extracting visible parts under heavy occlusion, demonstrating strong robustness. The resulting high-quality instance masks not only enhance the model's spatial perception of occluded objects but also provide more accurate and structurally coherent inputs for training the subsequent restoration model, SwingNet.

B. Amodal Mask Completion Experiment

The completion model is the second-stage processing model following instance segmentation, where the segmented masks

and RGB images are input to the model to predict the true object masks (amodal). In this experiment, we evaluate three different models along with our proposed model, with and without the COT-loss strategy. The PCNET-M [7] model is a self-supervised approach, which does not require the use of ground truth labels during training. However, this leads to a dependence on occlusion relation maps during inference and introduces some distortion in the predicted amodal masks. In practical scenarios, we typically only have RGB images available. The ASBU [30] model is a weakly supervised approach that also relies on occlusion boundaries to predict the true object masks. BCNet [44] is a fully supervised method that models occlusion through a bilayer decoupling structure, explicitly separating occluding and occluded instances. In contrast, SwingNet requires only the RGB image of the object and the segmented visible mask (modal) to predict the true shape of the object (amodal). We acknowledge that the compared methods employ different levels of supervision (self-supervised, weakly supervised, and fully supervised). This comparison is intentionally designed to be comprehensive: (1) PCNet-m and ASBU represent the dominant paradigms in the amodal completion literature and are the most widely cited baselines in this field; (2) BCNet [44] is included as a fully supervised baseline to provide a fair same-supervision-level comparison; (3) from a practical application standpoint, what ultimately matters is the final restoration quality regardless of the training paradigm—our industrial sorting scenario requires the best possible mask completion accuracy, and the comparison demonstrates that our fully supervised approach with dedicated architecture design achieves superior results. Furthermore, the performance gap between SwingNet and BCNet (both fully supervised) validates that our architectural innovations (cascaded Unet-blocks with attention mechanisms) and COT-Loss strategy contribute meaningfully beyond the supervision advantage alone. The experimental results are shown in Table III, and Fig. 6 illustrates the amodal mask completion performance of SwingNet on the KWOD dataset.

The test results of four models in the restoration experiment: PCNet-m, ASBU [30], SwingNet, and SwingNet + COT-Loss on COCOA, KINS, and KWOD are shown in Table III. The results for PCNet-m and ASBU on the COCOA and KINS datasets are referenced from [30]. The mean Intersection over Union (mIoU) measures the overall performance of the model from modal mask to amodal mask, while the Intersection over Union for the invisible parts (inv-mIoU) reflects the model's restoration performance. In the experiments shown in Table III, SwingNet outperforms PCNet-m and ASBU on most metrics across the three datasets, particularly demonstrating significant improvements in the intersection over union of the invisible regions (inv-mIoU). This indicates that SwingNet is better adapted for the mask completion task, with a notable performance boost of 17.53% on the KINS-test dataset. Furthermore, incorporating the loss strategy into SwingNet further improves the model's performance, with increases in both mIoU and inv-mIoU ranging from 0.1% to 1%. Unfortunately, SwingNet seems less adapted to the KINS dataset, resulting in slightly lower mIoU scores compared to the first two models. However, with the addition of the loss strategy, SwingNet achieves

an impressive 96.6% performance on our self-built KWOD dataset. Although the inv-mIoU on KWOD does not reach the same level of mask completion performance as seen on KINS-test, the lower performance on KWOD can be attributed to the presence of not only partial occlusions but also split occlusions (as addressed in Section 3.2, Instance Segmentation). This increased complexity in occlusion further challenges the mask completion task.

Moreover, the experimental results indicate that the proposed SwingNet, along with the COT-Loss-enhanced two-stage architecture, demonstrates strong generalization and adaptability in the task of occluded object mask restoration. Traditional approaches often simplify occlusion recovery into a single end-to-end learning problem, lacking explicit structural modeling of occlusion relationships. This typically results in blurry or structurally inconsistent reconstructions of the invisible regions. In contrast, our method first employs high-quality instance segmentation to obtain accurate visible region information, and then utilizes a lightweight and structure-sensitive SwingNet to complete the occluded parts. This approach achieves a better balance between spatial continuity and semantic consistency. Additionally, the proposed COT-Loss guides the model to focus more effectively on discriminative features in the invisible regions, thereby enhancing the representation of complex occlusion relationships. Even under highly complex and realistic occlusion scenarios, our method consistently maintains superior performance, further validating the necessity and effectiveness of the proposed segmentation-to-restoration two-stage framework.

As shown in Fig. 7, in the actual conveyor belt scenario with garbage occlusion, the robot's grasping sequence and pose (such as grasping angle) are determined based on our proposed solution. It can be observed that, according to the occlusion sequence provided by the mask completion model, the robot can easily sort the occluded garbage items by following the correct order of occlusion.

TABLE IV
MODEL ABLATION DESIGN. THE HIGHLIGHTED PARTS ARE USED FOR THE FINAL MODEL.

Experiment	Method	Param(M)	Flops(B)	KWOD-val	
				mIoU	Inv-mIoU
Stage	One	27.42	88.37	96.58	65.25
	Two	30.51	120.24	96.67	74.31
Instance Model	Normal	-	-	92.23	68.52
	MD	-	-	96.67	74.31
Recover model	L = 1	1.55	11.51	95.78	70.25
	L = 2	3.11	22.29	96.56	72.52
	L = 3	4.65	33.13	96.67	74.31
Loss strategy	Fixed	-	-	96.31	69.70
	Varietal	-	-	96.67	74.31

C. Ablation Study

As shown in Table IV, we performed an ablation study on the core components of SwingNet to assess their necessity. Following the one-variable-at-a-time principle, all variant models were trained on the same dataset as our proposed main model and evaluated on the KWOD validation set.

Stage. In the single-stage multi-head model—widely adopted in many applications—we drew inspiration from clas-



Fig. 7. Robot grasping sequence and grasping angles obtained through the proposed solution.

sic instance-segmentation architectures (YOLOv5, YOLOv8) by adding an extra amodal detection head. For each object, we multiply its k amodal coefficients (confidence scores) by the corresponding k amodal prototypes, sum the products, and then apply a sigmoid nonlinearity (σ) to produce the final amodal mask. This is formalized as:

$$M_a = \sigma(PC^T) \quad (8)$$

where $P \in \mathbb{R}^{k \times H \times W}$ denotes the k amodal prototypes and $C \in \mathbb{R}^k$ represents the k amodal coefficients.

The two-stage model builds on the single-stage backbone (without the amodal head) by appending a recovery network (SwingNet). Experimental results demonstrate that, while the single-stage multi-head design yields higher runtime efficiency, it lags behind the two-stage approach in accuracy—particularly when predicting the occluded regions of objects.

Instance Model. In Table IV, “Normal” refers to the standard instance segmentation model without any modification, while “MD” stands for Mask Disentanglement—our proposed strategy that explicitly segments both visible flanks of an object whose central region is occluded, treating them as a single instance rather than separate objects (as described in Section 3.2). Our instance segmentation differs from standard approaches in that it explicitly segments both visible flanks of an object whose central region is occluded, thereby harvesting sufficient visible cues to support the downstream recovery model. Empirical results show not only superior performance on the visible portions but also a 5% improvement in the restoration of the occluded regions. This indicates that the

proposed model is more suitable for the mask completion task than the normal model, and the mask disentanglement strategy further improves the model’s performance.

Recovery Model. The restoration model we propose is SwingNet (see Sec. 3.B), whose core building block is a streamlined U-Net module. In Table IV, L denotes the number of such modules—and hence the overall network depth—so that as L increases, the parameter count and training time grow accordingly, and optimization becomes more difficult.

Loss Strategy. This loss-function strategy is tailored to the training of the recovery model. After the model has sufficiently learned the visible regions—i.e. when its mIoU on visible areas exceeds 80%—we introduce the Invisible Loss, i.e. the region-restricted cross-entropy term defined in Eq. 2 that is focused exclusively on the occluded regions. This encourages the network to devote more capacity to reconstructing invisible parts. Experimental results show that, while overall mIoU drops marginally, the mIoU on occluded regions increases substantially.

TABLE V
EXPERIMENTAL RESULTS OF ROBOTS SORTING FIVE TYPES OF GARBAGE UNDER DIFFERENT ALGORITHMS.

Model\class	Plastic bottle	Plastic box	Metal bottle	Glass bottle	Paper box	SSR
Occurrences	65	33	21	47	45	-
Sort with Detection	57	19	16	36	38	78.67%
Sort with OSRNet	60	26	19	34	42	85.78%

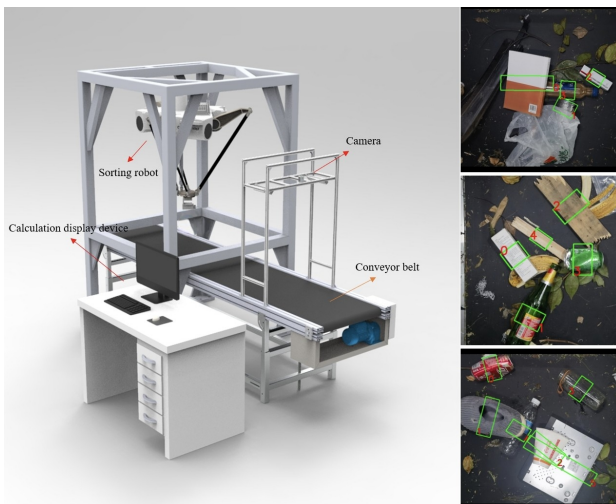


Fig. 8. Kitchen waste sorting system. The system's front section houses the computation and monitoring hardware, while the rear section comprises the robotic sorting apparatus, which consists of a camera, a conveyor belt, and the sorting robot.

D. Application Evaluation

We evaluated OSRNet through a pilot deployment on a real-world kitchen-waste sorting robot. As shown in Fig. 8, the hardware comprises a computing unit and a robotic sorting arm. The computing unit performs inference on images captured by the camera to predict grasping-box coordinates in the image plane. Through coordinate transformation—established via hand-eye calibration—it translates the box's image coordinates, grasping sequence, and related data to the robot controller. When the conveyor belt transports the target item into the arm's reachable workspace, the robotic arm executes the sorting operation.

To mirror real-world kitchen-waste scenarios, we mixed recyclable and non-recyclable items on the conveyor belt and ran it at a constant speed while simulating the gripper's pick actions. An object was marked as “unsorted” if it passed beyond the gripper's reachable envelope without the gripper having attempted a pick directly overhead. The results—summarized in Table V—focus on the five most populous categories, comparing our method against a standard object-detection baseline (YOLOV8 [14]).

Our OSRNet outperforms the detection model in multiple categories, yielding an overall sorting success rate (SSR) roughly 7% higher. This gain arises because the gripper can more reliably grasp objects in stacked or overlapping arrangements. As expected, however, OSRNet's performance on glass bottles trails the detection baseline: many bottles are transparent and lack affixed labels, making them harder to detect, and the mask-based reconstruction may introduce slight localization errors in the predicted grasp boxes.

We also conducted experiments on special occlusion scenarios, as shown in Fig. 9, involving three types of challenging cases: Intra-class Occlusion, Partial Embedding Occlusion, and Dual-End Residual Occlusion. Intra-class Occlusion occurs when two objects of the same category and similar appearance partially block one another. Partial Embedding

Occlusion refers to a situation in which part of the target object is enveloped by another object. Dual-End Residual Occlusion arises when two similar objects obscure opposite halves of each other—one concealing the upper half of the other's mask while the second occludes its lower half—often causing them to appear as a single merged object. As expected, Intra-class Occlusion is relatively straightforward, and our model can handle it effectively. However, Partial Embedding and Dual-End Residual Occlusions present greater challenges, despite their rarity in real-world scenarios. While our approach meets basic occlusion-handling requirements, in more complex environments the restoration model tends to “fill in” mask gaps, making it difficult to recover each object's true, complete contour.

E. Limitation Discussion

Although the proposed OSRNet achieves promising results in occluded object restoration and sorting, several limitations remain that warrant further investigation:

Sensitivity to Severe Occlusion Ratios. When the occlusion ratio exceeds 70%, the restoration model's performance degrades noticeably. The limited visible cues available in such extreme cases make it challenging for SwingNet to infer the complete object boundary, often resulting in over-smoothed or geometrically distorted predictions. This is reflected in the gap between mIoU and inv-mIoU scores across all datasets.

Dependence on Instance Segmentation Quality. As a two-stage pipeline, errors from the first-stage instance segmentation propagate to the restoration model. When the segmentation model fails to correctly identify fragmented visible regions as belonging to the same object—particularly for thin or elongated items—the downstream restoration becomes unreliable.

Limited Generalization to Transparent and Textureless Objects. As observed in the glass bottle category (Table V), transparent or textureless objects pose a fundamental challenge. The RGB-based feature extraction struggles to capture meaningful appearance cues for such objects, leading to degraded mask predictions.

Feature Representation Analysis. We observed that the intermediate feature maps in SwingNet's successive Unet-blocks exhibit a progressive refinement pattern: early blocks primarily recover coarse shape outlines, while later blocks focus on fine boundary details. However, when the occluded region lacks strong contextual cues from neighboring visible pixels, the attention mechanism tends to distribute weights uniformly rather than focusing on informative regions, reducing its effectiveness.

Future Directions. To address these limitations, future work could explore: (1) incorporating depth information or multi-view cues to provide additional geometric constraints for severely occluded objects; (2) introducing shape priors learned from category-specific templates to regularize the restoration output; and (3) designing an end-to-end joint optimization framework that allows the restoration model to provide feedback to the segmentation stage, enabling iterative refinement.



Fig. 9. Special occlusion cases. The first three columns present the instance-segmentation results, and the last three columns show the restoration-model outputs.

V. CONCLUSION

This paper presents a comprehensive solution for the challenging task of garbage sorting in scenarios with severe occlusion. Firstly, we established the Kitchen Waste Occlusion Dataset (KWOD), which provides data-driven support for model training. Secondly, in the segmentation-restoration model, we addressed the issue of visible parts of occluded objects being fragmented and misclassified as multiple objects in the instance segmentation module. In the second part, the mask restoration module, we redesigned the residual block and loss strategy, introducing a novel lightweight neural network, SwingNet. This network demonstrates optimal performance in mask restoration (specifically in inv-miou) on our self-built KWOD dataset, as well as the public COCOA and KINS datasets. Finally, by inferring the occlusion relationships between objects from the segmentation-restoration results, the robot is able to sort objects efficiently. The complete mask map also generates precise grasping boxes, enabling the robot to perform accurate object grasping.

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